

# The Silent Tax: How Typos Reshape Reasoning in Thinking LLMs

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**How changes are marked.** ~~Red strikethrough~~: deleted. Blue underline: added in the ACL-format, writing and bibliography review. Green underline: added because of the results checks (new numbers from the fixed code and repaired data, regenerated tables and Figure 1, text corrected to match the data). Figures are images and show the newer version. This file shows every change from the team’s version (19 Sep) to the submitted version.

## Abstract

Reasoning models perform strongly on clean mathematical and scientific benchmarks, but real users often write with typos. We study how these everyday errors affect not only final-answer accuracy, but also the reasoning process itself.

We introduce a controlled typo generator that independently varies the number of corrupted words and the proportion of typos that produce another valid English word. ~~We~~, apply it to 500 questions from each of GSM8K, MATH-500, and ARC-Challenge, and evaluate DeepSeek-R1-Distill-Qwen-7B across all corruption settings.

~~Typos consistently~~ Typos reduce accuracy, ~~with the largest effect~~ most on GSM8K, where ~~performance falls from 91.4~~ it falls from 92.6% on clean questions to ~~60.464~~ 60.5% in the most corrupted setting. ~~More importantly, typos~~ Typos that produce another valid ~~English word~~ are consistently word are more harmful than non-word typos. ~~Depending on the dataset,~~ a real-word typo is estimated to be ~~1.9 to 5.1~~ 1.3 to 1.7 times more damaging per typo.

on MATH-500 and GSM8K, and on ARC-Challenge only real-word typos have a detectable effect. The reasoning traces show that the model often notices that something is wrong: ~~corrupted inputs lead to,~~ with longer reasoning and more uncertainty language. ~~However,~~ but this does not translate into more effective self-correction. ~~The:~~ the model hesitates more, but ~~does not verify more.~~

~~Finally, none verifies only slightly more.~~ None of the lightweight ~~mitigation strategies~~

~~we tested consistently improved performance.~~ ~~Warning the model does not improve performance, asking it to rewrite~~ mitigations we tested on GSM8K (a typo warning, rewriting the question first ~~can make performance worse, and spell checking becomes less effective as the proportion of real-word typos increases.~~ Overall, our results ~~suggest that the,~~ or an external spell checker) consistently improved accuracy. The main challenge is not ~~only~~ detecting that the input is corrupted, but ~~correctly recovering the~~ recovering its intended meaning.

## 1 Introduction

Chain-of-thought prompting and reasoning-focused models have substantially improved performance on mathematical and scientific benchmarks (Wei et al., 2022; Guo et al., 2025). However, benchmark questions are ~~usually~~ carefully written, while real ~~user inputs often contain typos~~ users often make simple typing mistakes such as swapped letters, missing characters, or nearby-key ~~mistakes.~~

~~This gap motivates our study: real users often make simple typing mistakes when interacting with LLMs, so strong errors.~~ Strong performance on clean benchmark text may ~~therefore~~ overestimate reliability in real-world use.

Previous work has shown that typos can hurt LLM performance ~~,~~ often linking this effect to subword tokenization (Gan et al., 2024; Chai et al., 2024; Zhao et al., 2026) (Gan et al., 2024; Zhao et al., 2026), partly because they disrupt subword tokenization (Chai et al., 2024). However, these studies mainly focus on final-answer accuracy and do not examine what happens inside the reasoning process.

Reasoning models make this question especially interesting. Their long reasoning traces give them many opportunities to notice a corrupted word, infer what was intended, and recover before pro-

ducing the final answer. Do they actually do so? ~~We study this question using a controlled two-axis experimental design.~~

To study this, our typo generator separates two factors: how many words are corrupted and what kind of corruption they contain. In particular, we vary the share of typos that ~~accidentally~~ produce another valid English word. This allows us to compare real-word and non-word typos while keeping the total number of typos fixed and using the same questions in every condition—, a comparison that ~~previous work has~~, to our knowledge, previous LLM typo studies have not made.

We evaluate DeepSeek-R1-Distill-Qwen-7B (Guo et al., 2025) across the full set of conditions on three datasets with different reasoning characteristics. ~~Beyond measuring final answer accuracy, we also~~, and examine the reasoning traces themselves using word-level alignment and an LLM judge.

Our main contributions are: (i) a controlled experimental setup that independently varies typo rate and the proportion of real-word typos across three datasets, ~~allowing us to directly compare real-word and non-word corruptions while keeping the overall typo rate fixed.~~; (ii) an analysis of how typos affect the reasoning process itself, including ~~changes in~~ model uncertainty, whether the model notices corrupted words, and whether it ~~attempts to repair them, repairs them;~~ and (iii) an evaluation of three simple mitigation strategies: warning the model that the input may contain typos, asking it to correct possible typos before solving the problem, and applying an external ~~spellchecker before sending the prompt to spell checker before the model. model sees the question.~~

Our main findings are as follows. Typos cost up to 28.1 percentage points of accuracy on GSM8K, and the loss comes from wrong answers rather than truncated generations (§5.1). At a fixed number of typos, real-word typos are more harmful than non-word ones, by 1.3–1.7× per typo on MATH-500 and GSM8K, and on ARC only real-word typos have a detectable effect (§5.2). Contrary to our initial expectation that real-word typos would pass unnoticed, the model flags corrupted words, and hedges, more often as corruption grows, yet it also misreads more words and verifies only slightly more (§5.3, §5.4). Finally, none of the three mitigations reliably recovers the loss (§5.5).

**Clean** (model answers 18, correct)

*Janet’s ducks lay 16 eggs per day. She eats three for breakfast every morning and bakes muffins for her friends every day with four. She sells the remainder at the farmers’ market daily for \$2 per fresh duck egg. How much in dollars does she make every day at the farmers’ market?*

**25% typos, 40% of them real words**  $r=25\%$ ,  $\rho=40\%$   
(model answers 2624, wrong)

*Janet’s ~~duck~~ lay 16 eggs per ~~adgy~~. ~~Sje~~ ~~east~~ ~~rthere~~ for breakfast every morning and bakes muffins for ~~uher~~ ~~friend~~ every ~~azy~~ with four. She sells the remainder at the farmers’ market daily for \$2 per ~~resh~~ duck egg. How much in dollars does she make every day at the ~~faremrs~~’ market?*  
eats→east——ducks→duck——three→rthere  
friends→friend——fresh→resh——her→uher  
*Janet’s duck lay 16 eggs per adgy. Sje east rthere for breakfast every morning and bakes muffins for uher friend every azy with four. She sells the remainder at the farmers’ market daily for \$2 per resh duck egg. How much in dollars does she make every day at the faremrs’ market?*

Figure 1: A single GSM8K item under our generator. Numbers and  $\LaTeX$  are protected, ~~and~~; only alphabetic words are corrupted. ~~The two real-word~~ Four of the ten typos are real words (ducks→duck, eats→east, friends→friend, fresh→resh); three of them (ducks→duck, eats→east, friends→friend) are invisible to a spell checker of \$5.5. The model ~~loses the “three eaten” quantity and~~ answers 2624 instead of 18.

## 2 Background and Related Work

**Sensitivity to character-level typos.** Belinkov and Bisk (2018) show that both synthetic and natural character-level noise ~~substantially~~ degrade neural machine translation, and that robustness to one type of noise does not necessarily transfer to others. ~~Later work studied similar vulnerabilities in adversarial settings.~~ HotFlip (Ebrahimi et al., 2018) shows that a few white-box character changes can alter a classifier’s prediction, while TextFooler (Jin et al., 2020) generates black-box adversarial examples through word substitutions. PromptRobust (Zhu et al., 2024) shows that small changes to task prompts can substantially reduce LLM performance, including changes at both the character and word levels. Pruthi et al. (2019) defend against adversarial misspellings by placing a robust Small adversarial character- and word-level edits also fool text classifiers (Ebrahimi et al., 2018; Jin et al., 2020) and degrade LLMs through their prompts (Zhu et al., 2024). Two lines of defense mirror

our mitigations: Pruthi et al. (2019) place a word-recognition model before the classifier. In one experiment, an adversarial character attack reduced BERT accuracy from 90.3% to 45.8%, while their defense recovered it to 75%. This approach is closely related to the spell-checking mitigation we evaluate. In front of the classifier to undo adversarial misspellings, and Wang et al. (2024) find that having an LLM clean a noisy instruction before the task is difficult, particularly for open-source models.

**Typos and modern LLMs.** Recent studies show that typos can substantially reduce LLM performance. Gan et al. (2024) use an optimized adversarial search to inject typos into reasoning benchmarks such as GSM8K, BBH, and MMLU, and show that model accuracy drops substantially as the number of character-level errors increases. Chai et al. (2024) show that character-level typos can hurt model performance partly because they change how words are split into subwords. Zhao et al. (2026) study typo robustness across several 12 languages and 18 open-source models. They and find that typos hurt performance most on tasks that require generation or more complex reasoning. Concurrently, Hu et al. (2026) report that a thinking budget recovers most of the accuracy lost to QWERTY keyboard perturbations, and trace the harm to how many of the question’s tokens are destroyed.

These studies mainly measure final-answer accuracy rather than the reasoning process itself, and they vary typos by generation mechanism rather than by whether the corrupted form is itself a valid word. That distinction matters is classic in spelling correction, where real-word errors cannot be caught by dictionary lookup (Kukich, 1992; Hirst and Budanitsky, 2005), and it matters here too. A typo that creates a non-word, such as triangle→trianlge, mainly affects tokenization, while a typo that creates another valid word, such as sum→sun, leaves tokenization intact and is tokenized normally but changes the meaning of the sentence. Our study separates these cases and examines how the model responds to each during reasoning.

**Reasoning traces and self-correction.** Reasoning models such as DeepSeek-R1 (Guo et al., 2025) produce long reasoning traces before giving a final answer, allowing us to examine how they respond to errors during reasoning. However, these. These

traces do not always fully reflect the model’s internal reasoning (Turpin et al., 2023), and prior work shows that LLMs often struggle to correct their own reasoning without external feedback, sometimes even degrading their answers in the process (Huang et al., 2024).

Our setting asks a related but different question: can a long reasoning process help the model deal with errors that are already present in the input? A typo may be detectable from context, giving the model an opportunity to notice it, infer the intended word, and continue reasoning correctly. This allows us to study whether additional reasoning actually helps models recover from noisy input.

### 3 Method

#### 3.1 Controlled typo generation

We generate corrupted versions of each question by modifying only its natural-language text, building on the keyboard-aware edit operations of MulTypo (Zhao et al., 2026) and adding control over real-word typos. Numbers,  $\LaTeX$  expressions, and  $\LaTeX$  command names are protected and never modified. Candidate words must consist only of ASCII letters, and we exclude single-character words and a small set of common function words.

The generator varies two factors independently. The typo rate  $r$  determines how many eligible words are corrupted, while the real-word ratio  $\rho$  determines the target fraction of those corruptions that should result in another valid English word. For example, eats → east is a real-word typo, while ducks → dcks is a non-word typo. Words are selected for corruption before  $\rho$  is applied, so for a fixed typo rate the number of corrupted words is identical across all values of  $\rho$ .

Each corrupted word non-word typo receives a character-level edit sampled with weights 0.40 adjacent-key substitution, 0.20 deletion, 0.20 insertion, and 0.20 transposition; a real-word typo is drawn from the valid words that these edits can reach, weighted by the same probabilities. With probability 0.10, a second edit is applied to the same word.

For each question, we convert  $\rho$  into a target number of real-word typos, using probabilistic rounding when the target is not an integer. We then process the selected words in random order. While the target has not been reached, the generator searches for valid real-word variants using the NLTK English lexicon (Bird et al., 2009). If none

exists for a selected word, that word receives a non-word typo and the generator continues trying to satisfy the target with the remaining words. Any real word produced accidentally is counted toward the target. Because the selected words are not chosen based on whether a real-word variant exists, the requested ratio cannot always be reached exactly. In these cases, the generator gets as close to the target as the selected words allow. Table 1 reports the achieved ratios.

For each benchmark, we create a clean condition that is identical to the original dataset, together with nine corrupted conditions. These combine typo rates  $r \in \{25, 50, 75\}\%$  with real-word ratios  $\rho \in \{10, 40, 70\}\%$ . The generator is seeded consistently across conditions, and the same questions are used in every condition, allowing direct paired comparisons. The resulting dataset variants are available on the Hugging Face Hub.

### 3.2 Datasets

We use 500 questions from each of three benchmarks: GSM8K (Cobbe et al., 2021), MATH-500 (Hendrycks et al., 2021; Lightman et al., 2024), and ARC-Challenge (Clark et al., 2018). We chose these datasets because they differ in how much of the information needed to solve a question is expressed in natural-language text.

GSM8K contains arithmetic word problems in which most of the problem whose structure is described in prose. Its questions are also relatively long, so more words are available for corruption almost entirely in prose, and its questions are relatively long. MATH-500 contains competition mathematics problems that are shorter and contain shorter competition problems with substantial L<sup>A</sup>T<sub>E</sub>X content, which is protected by our generator our generator protects. ARC-Challenge contains shorter multiple-choice science questions, where the answer options provide additional information that whose answer options may help the model interpret a corrupted question.

These differences mean that As a result, the same typo rate does not produce the same absolute number of typos across datasets. GSM8K receives the most corruption per question, while MATH-500 and ARC-Challenge receive fewer corruptions because they contain fewer eligible words (Table 1). We therefore compare corruption levels within each dataset rather than treating a given typo rate as the same absolute amount of noise across all three benchmarks. Table 1

GSM8K-MATH-500-ARC-C questions 500-500-500 mean words per question 45.8-31.5-23.5 median words per question 43-26-20 eligible words per question 30.6-15.7-16.6 typos per question,  $r=25\%$  7.6-4.0-4.2 typos per question,  $r=50\%$  15.3-7.9-8.3 typos per question,  $r=75\%$  22.9-11.8-12.4 achieved real-word ratio,  $\rho=10\%$  0.100-0.099-0.102 achieved real-word ratio,  $\rho=40\%$  0.399-0.397-0.401 achieved real-word ratio,  $\rho=70\%$  0.698-0.681-0.682 total corrupted words 68,739-35,685-37,248

|                                       | GSM8K  | MATH-500 | ARC-C  |
|---------------------------------------|--------|----------|--------|
| questions                             | 500    | 500      | 500    |
| mean words per question               | 45.8   | 31.5     | 23.5   |
| median words per question             | 43     | 26       | 20     |
| eligible words per question           | 30.6   | 15.7     | 16.6   |
| typos per question, $r=25\%$          | 7.6    | 4.0      | 4.2    |
| typos per question, $r=50\%$          | 15.3   | 7.9      | 8.3    |
| typos per question, $r=75\%$          | 22.9   | 11.8     | 12.4   |
| achieved real-word ratio, $\rho=10\%$ | 0.100  | 0.099    | 0.102  |
| achieved real-word ratio, $\rho=40\%$ | 0.399  | 0.397    | 0.401  |
| achieved real-word ratio, $\rho=70\%$ | 0.698  | 0.681    | 0.682  |
| total corrupted words                 | 68,739 | 35,685   | 37,248 |

Table 1: Dataset and corruption statistics over the nine corrupted conditions ( $r \in \{25, 50, 75\}\%$ ,  $\rho \in \{10, 40, 70\}\%$ ), on the first 500 questions per benchmark of GSM8K, the first 500 four-option questions of ARC-Challenge, and all of MATH-500. Eligible words are those the generator may corrupt, after protected spans, single-character words and function words are removed; the typo rate applies to them, and real-word ratios are pooled over all corrupted words.

reports the resulting corruption statistics across benchmarks.

## 4 Experimental Setup

**Models.** We use DeepSeek-R1-Distill-Qwen-7B (Guo et al., 2025) as our main model and access it, accessed through the Hugging Face API. We run it, with temperature 0.6 and top- $p$  0.95, using the default settings from the model card, and generate one response per question. We use the recommended decoding settings to better (the model-card defaults, to reflect how a typical user would interact with the model, rather than tuning them specifically for our evaluation. For each dataset, we set the run it) and one response per question. The maximum generation length to a value that is 20,000 tokens for GSM8K and 17,000 for MATH-500 and ARC-Challenge, which we found sufficient for the model to produce complete answers for a large enough portion of the questions.<sup>1</sup>

For the LLM-based evaluation, finish on a large majority of questions. For the LLM judge of the

<sup>1</sup>In practice this is 20,000 generated tokens for GSM8K and 17,000 for MATH-500 and ARC-Challenge.

reasoning traces we use Llama-3.3-70B-Instruct (Grattafiori et al., 2024), also through the Hugging Face API, with temperature 0.

**Prompts.** For math problems, we provide the question followed by the instruction: “Please reason step by step, and put your final answer within `\boxed{}`.” For multiple-choice questions, we present the four answer options with shuffled labels and ask the model to return the final answer as a boxed letter. Typos are added only to the question itself, while the instruction remains unchanged and typo-free.

**Scoring.** We extract the final answer only from the text produced after the closing `</think>` tag. Responses ~~that without an extractable final answer, almost always because they~~ reach the token limit ~~without producing a final answer~~, are marked as unanswered. GSM8K is evaluated using normalized numeric matching, MATH-500 using symbolic equivalence with `math_verify`, and ARC using exact answer-letter matching. We report accuracy over answered questions ~~and separately report together with~~ the number of answered responses, ~~and check the effect of truncation with strict accuracy, which counts unanswered responses as wrong.~~

**Statistics.** Since each condition contains the same 500 questions, we use paired ~~statistical~~ tests. We ~~report the number of~~ count the questions that change from correct to incorrect and from incorrect to correct, and ~~use a continuity-corrected McNemar test to compare these paired outcomes~~. ~~This analysis helps separate with a continuity-corrected McNemar test, which separates~~ the effect of typos from differences in question difficulty, ~~since each comparison is made on the same set of questions. Tables report uncorrected  $p$ -values; in the text we also state which results survive a Holm correction over the nine tests of each family within a dataset.~~

**Reasoning trace analysis.** We analyze the ~~model’s~~ reasoning traces in three ways. First, ~~we use word-level alignment to track tracks~~ how the model handles each corrupted word. ~~We check of three or more letters:~~ whether the trace uses the original word, the corrupted ~~version, or an explicit correction, and classify form, or both, classifying~~ each typo as silently fixed, flagged, misread, or not used.

~~(§5.3). Second, we measure self-doubt directly from the reasoning traces. We define sets of words and phrases associated with uncertainty and self-correction, and count how often they appear in each trace. Since traces differ in length, we normalize these counts count curated uncertainty and second-guessing expressions, normalized by trace length. We then compare these measures across typo conditions and between correct and incorrect answers to examine whether higher uncertainty is associated with model failure.~~

~~Third, we use (§5.4). Third,~~ an LLM judge. ~~The judge receives the clean question, the corrupted question (Zheng et al., 2023) receives the list of corrupted words with their originals (at most 25), the question as shown to the model (for ARC, without the answer options), the list of changed words, and the model’s reasoning trace reasoning trace (its first and last 2,000 characters). Using a predefined evaluation prompt fixed rubric, it assigns a self-doubt self-doubt score from 0 (a clear, straight-line solution) to 10, which measures the “amount of unambiguity” and a repair behavior (pervasive uncertainty, reversals or looping), and a repair score from 0 (the intended meaning of the question is fully understood) to 5 which measures how much the model understands the real meaning of the question. (the model reasons from the wrong meaning). Higher is worse on both scales. The judge scores answered traces only.~~

## 5 Results

### 5.1 Effect of typos on accuracy

Figure 2 and Table A1 in the appendix show the full results. On GSM8K, answered-only accuracy drops from ~~91.4~~92.6% on clean questions to ~~60.4~~64.5% in the most corrupted condition ( $r=75$ ,  $\rho=70$ ), a decrease of ~~31.0~~ percent. ARC shows a smaller drop from ~~84.7~~ to ~~74.7~~%, while ~~28.1~~ percentage points. ARC drops from ~~85.2~~ to ~~74.5~~%, and MATH-500 drops from ~~95.2~~ to ~~86.9~~%. Overall, accuracy decreases as the typo rate increases across all three datasets (Figure 2). On all three datasets accuracy is lowest at the highest typo rate, although on ARC the mildest rate ( $r=25$ ) is still on par with clean.

The paired analysis shows the same trend. In the most corrupted GSM8K condition, ~~164~~148 questions change from correct to incorrect, while only ~~14~~11 change from incorrect to correct, out

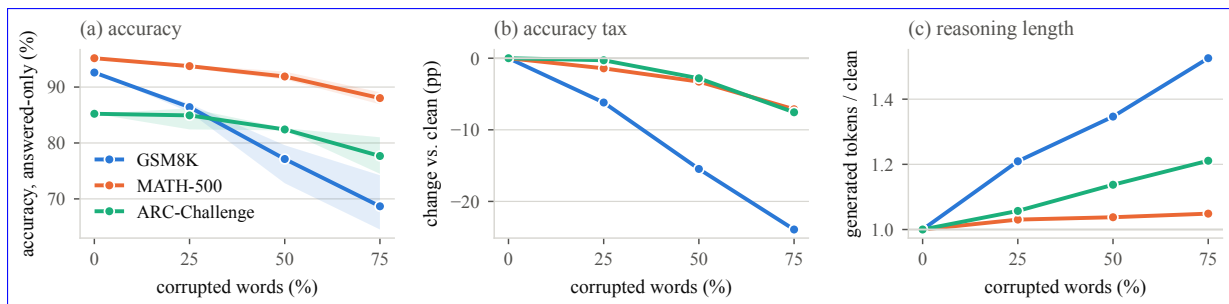


Figure 2: **The accuracy tax.** (a) answered-only accuracy against the typo rate; the line is the mean over the three real-word ratios and the band spans them. (b) the same as a change from clean, in percentage points. (c) mean generated tokens relative to clean. Accuracy falls with the typo rate on all three datasets and falls fastest on GSM8K, where the text carries the whole problem and each question receives the most typos. On average, reasoning gets longer under typos, so the loss is not the model giving up early.

of 480-482 questions answered in both conditions ( $p < 10^{-6}$ ). ARC shows 75-77 correct-to-incorrect versus 26-25 incorrect-to-correct changes, and MATH-500 shows 46 versus 7, both statistically significant. After Holm correction, all nine GSM8K is affected even at the mildest typo condition, while significant effects appear only at higher typo rates for conditions remain significant, against four on MATH-500 and ARC (Table A1).

One possible explanation for these differences (all at  $r > 50$ ) and two on ARC (both at  $r=75$ ). Part of the gap may reflect the larger number of typos per GSM8K question at the same rate (Table 1); another possible explanation is how much each task depends on the wording of the question. GSM8K problems rely heavily on the text to describe the quantities and relationships needed for the solution. ARC provides multiple-choice options that, ARC's answer options may help the model recover some information, while and much of the important mathematical content in mathematical content of MATH-500 is written in protected L<sup>A</sup>T<sub>E</sub>X and is therefore not affected by our typo generator.

Typos also lead to longer reasoning traces across all three datasets, while the model still produces final answers in most cases. On Truncation does not explain the loss. Counting unanswered responses as wrong, GSM8K, for example, the average reasoning length increases accuracy falls from 92.6% to 62.2%; of this strict drop at  $r=75$ ,  $\rho=70$ , 92% comes from lower accuracy on answered responses and only 8% from truncation; on MATH-500 truncation contributes nothing and on ARC 5%. Typos also lengthen the reasoning: on GSM8K the mean generated length grows from 1,582-535 tokens on

clean questions to 2,594 tokens-330 in the most corrupted condition (and up to 2,593 at  $r=75$ ,  $\rho=40$ ), with smaller increases on MATH-500 and ARC. This suggests that the drop in accuracy is driven mainly by incorrect answers rather than incomplete generations. In fact, wrong answers tend to have longer reasoning. Wrong answers have longer traces than correct ones, indicating that in every configuration, so additional reasoning is more closely associated with difficulty rather than with successful recovery (Figure 2c; Table A4).

**The accuracy tax.** (a) answered-only accuracy against the typo rate; the line is the mean over the three real-word ratios and the band spans them. (b) the same as a change from clean, in percentage points. (c) mean generated tokens relative to clean. Accuracy falls with the typo rate on all three datasets and falls fastest on GSM8K, where the text carries the whole problem. Reasoning gets longer under typos everywhere, so the loss is not the model giving up early.

## 5.2 Real-word typos are more harmful

Because typo rate and real-word ratio are varied independently, we can compare real-word and non-word typos while controlling for the total number of typos. For each dataset, we fit a logistic regression using the that predicts whether the final answer is correct from the number of real-word and non-word typos to predict whether the final answer is correct (Figure 3a).

The results show that an An additional real-word typo has a stronger negative association with correctness than an additional non-word typo. Based on the estimated odds multipliers from the logistic regression, the the relative per-typo loss is about  $1.91.7 \times$  larger on GSM8K, (95% CI 1.4-2.3) and

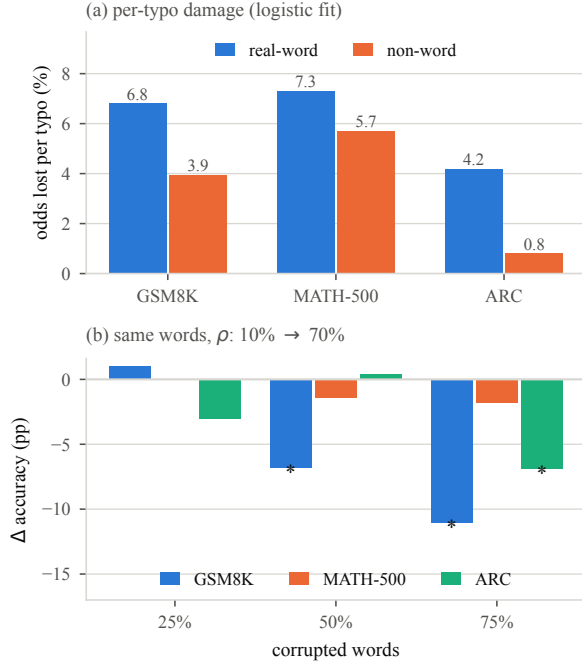


Figure 3: **The real-word effect.** (a) multiplicative loss in the odds of a correct answer per additional typo, from a logistic regression fitting real-word and non-word counts separately. (b) the controlled paired comparison: the same questions with the same number of typos, changing only the real-word share from  $\rho=10$  to  $\rho=70$ . \* marks McNemar  $p < 0.05$  (uncorrected).

1.3 $\times$  on MATH-500, and 5.3(1.0–1.7). On ARC the estimate is 5.2 $\times$  on ARC, but the non-word coefficient is not distinguishable from zero, so the ratio is not well defined (Table A2).

The paired comparisons support this pattern, although the effect depends on the overall typo rate. On GSM8K at a 75% typo rate, increasing the real-word share from 10% to 70% lowers accuracy by 14.1–11.0 percentage points, even though the same questions contain the same total number of typos. The difference is also significant at a 50% typo rate (both after Holm correction), but not at 25%. This suggests that when corruption is relatively mild, the model can often handle both real-word and non-word typos, while the difference between the two kinds of typo, and the difference becomes more pronounced as the overall typo rate increases (Figure 3b; Table A3).

ARC shows a significant real-word effect at the highest typo rate, while MATH-500 does not show a. On ARC, the effect is significant at  $r=75$  (–6.9 pp, surviving correction); one contrast at  $r=25$  ( $\rho=40 \rightarrow 70$ ) reaches  $p < 0.05$  but does not survive correction, and none at  $r=50$  does. MATH-500 shows no significant differ-

ence at any typo rate. This is consistent with the smaller number of corrupted words in MATH-500 questions, since much of their mathematical content is protected from modification. All nine contrasts, for all three datasets, are in Table A3.

Overall, the results suggest that real-word typos can be appear especially difficult for the model. Unlike non-word typos, which often produce visibly unusual word forms, real-word typos they result in valid words and can therefore change the meaning of the input without creating an obvious tokenization problem.

### 5.3 Repair behaviour : noticed but not always repaired

We recover each corrupted word by comparing the clean and corrupted versions of the question, and then classify how the model handles it in the reasoning trace:

**silent fix**: only the original word appears, suggesting that (the model recovered the intended word without explicitly mentioning the typo); **flagged**: both the original and corrupted forms appear, indicating that the corrupted form appear (the model noticed the typo and corrected it and corrected the typo); **misread**: only the corrupted form appears, suggesting that the model interpreted (the model took it as the wrong word); and **not used**: neither form appears in the reasoning trace. We leave out the words (about 3%) whose corrupted form also appears in the model’s trace on the clean question, since there its presence says nothing about the typo.

We originally expected real-word typos to go unnoticed more often, while non-word typos would be easier to detect and correct explicitly. The results only partly support this prediction.

Both a higher typo rate and a higher real-word ratio change the way the model handles corrupted words. As the typo rate increases from 25% to 75%, silent fixing decreases across on all three datasets. On: on GSM8K, for example, it falls from 73.374.0% to 55.8% at  $\rho=10$ . Similar decreases appear on ARC and MATH-500. At the same time, flagging and misreading become more common, and: pooled over real-word ratios, misreading roughly doubles on all three datasets, from 3.9% to 7.8% on GSM8K and from 1.4% to 2.7% on MATH-500, and rises less steeply on ARC, from 2.9% to 4.6% (Figure 4; Table A5).

Increasing the real-word ratio produces has a similar effect. Pooled across typo rates on GSM8K,

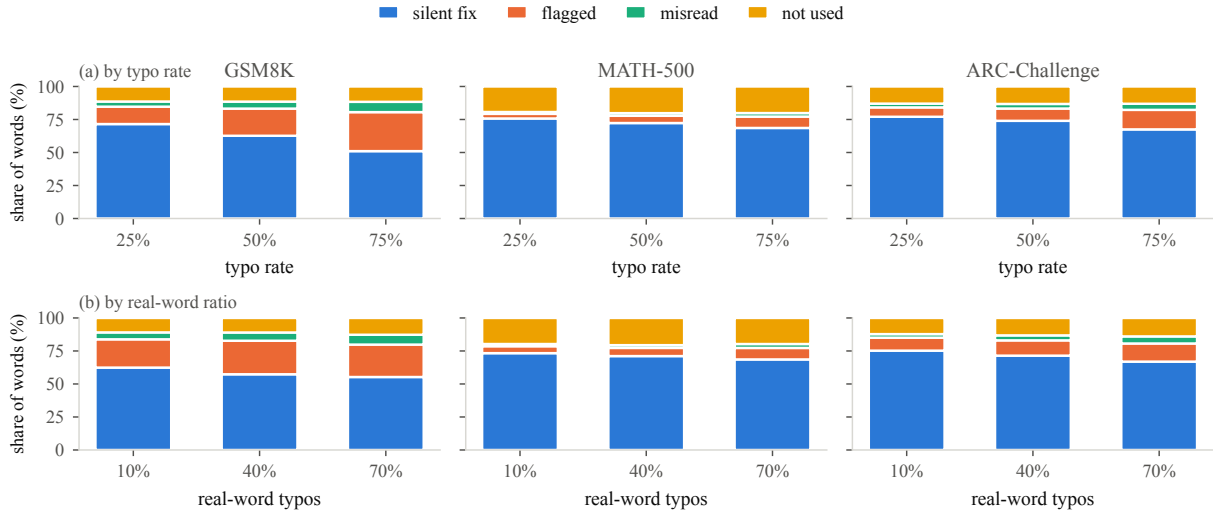


Figure 4: **Word-level repair behaviour.** How every each corrupted word (three or more letters, answered questions) is handled in the reasoning trace. (a) against the typo rate, pooled over real-word ratios. (b) against the real-word ratio, pooled over typo rates. Section 5.3 claims a trend on both axes and both are present: silent Silent fixing gives way to explicit flagging and to outright misreading as either knob is turned up the typo rate or the real-word ratio increases (on GSM8K, flagging levels off above  $\rho=40$ ), while the share of words never referred to at all stays roughly constant. Per-configuration values are in Table A5.

silent fixing falls from 62.3% to 52.9 62.4% to 55.3% as  $\rho$  increases from 10 to 70, while flagging rises from 21.0% to 26.8% 21.4% to 24.5% (peaking at 25.4% at  $\rho=40$ ) and misreading from 5.4% to 7.7% The 5.2% to 7.5%; the same overall pattern appears on MATH-500 and ARC. Misreading is also closely related to failure: it occurs in every configuration it is 1.2 to 2.0 times more often in incorrect answers than in correct ones frequent in GSM8K traces that end in a wrong answer, and 1.3 to 2.7 times on ARC; on MATH-500, where misreads are rare, the relation is not consistent (Table A5).

The proportion of typos classified as not used remains relatively The share of typos that are not used stays stable within each dataset, but differs across tasks: roughly 12% on GSM8K, 13% on ARC, and 19.20% on MATH-500. One possible explanation is that, possibly because MATH-500 reasoning relies more heavily on symbolic manipulation, so the model is more symbolic and refers back to individual words from the original the wording of the question less often (Figure 4).

In the LLM-as-a-judge analysis, a separate language model examined each reasoning trace and assigned a score for repair behaviour. The judgewas shown the corrupted words, the question, and the model’s reasoning trace. It then gave a score from 0 to 5: 0 means that the model fully understood the meaning of the question, while 5

means that it completely failed to understand the intended meaning. We then calculated the mean score for different typo rates, real-word ratios, and final answer outcomes.

The results show that more typos generally make the question harder to understand and repair. The average repair-understanding score increases as the typo rate rises, meaning that the model loses more of the original meaning when more words are corrupted. Wrong answers also receive higher average repair-understanding scores than correct answers, supporting the connection between misunderstanding. The judge’s repair score (§4; Table 2) tells the same story. It rises with the typo rate (on MATH-500 only at  $r=75$ ), and wrong answers score higher than correct ones (2.40 vs. 0.97 on GSM8K), linking misunderstanding of the corrupted question and failure. The to failure. It also rises with the real-word ratio has an additional effect: in every dataset and typo-rate cell, consistent with real-word typos look valid and can silently change the meaning of a sentence, making them harder to detect and repair than obviously invalid non-word typos (Table 2) typos being harder to repair, although we did not test this trend for significance.

## 5.4 Self-doubt

We measure self-doubt using with two curated sets of expressions found in the reasoning traces.

| condition                      | GSM8K  |       | MATH-500 |       | ARC-C  |       |
|--------------------------------|--------|-------|----------|-------|--------|-------|
|                                | repair | doubt | repair   | doubt | repair | doubt |
| clean                          |        |       |          |       |        |       |
| —                              | 0.12   | 4.46  | 0.03     | 3.66  | 0.03   | 4.70  |
| by typo rate $r$               |        |       |          |       |        |       |
| 25%                            | 0.88   | 4.86  | 0.35     | 3.90  | 0.97   | 5.12  |
| 50%                            | 1.27   | 5.12  | 0.35     | 3.98  | 1.03   | 5.31  |
| 75%                            | 1.74   | 5.67  | 0.48     | 4.01  | 1.35   | 5.63  |
| by real-word ratio $p$         |        |       |          |       |        |       |
| 10%                            | 1.13   | 5.20  | 0.32     | 3.87  | 0.94   | 5.33  |
| 40%                            | 1.32   | 5.29  | 0.39     | 4.03  | 1.10   | 5.28  |
| 70%                            | 1.43   | 5.15  | 0.47     | 3.99  | 1.31   | 5.45  |
| by final answer (typo configs) |        |       |          |       |        |       |
| correct                        | 0.97   | 4.75  | 0.30     | 3.78  | 0.98   | 5.01  |
| wrong                          | 2.40   | 6.83  | 1.36     | 5.86  | 1.76   | 6.92  |

Table 2: Mean LLM-judge scores over judged answered traces. *repair* is the repair score (0–5, higher means less of the intended meaning recovered); *doubt* is the self-doubt score (0–10, higher means more hesitation and looping).

The uncertainty set includes hedging expressions contains hedges such as *maybe*, *perhaps*, *possibly*, *might be*, and *not sure*. The ; the second-guessing set captures explicit reconsideration or self-correction, with expressions such as *wait*, *actually*, *reconsider*, *let me recheck*, and *double-check*. To account for differences in trace length, we report the number of matches. We count each expression once, even when several patterns match it, and report counts per 1,000 reasoning words.

Uncertainty increases as the input becomes more corrupted. On GSM8K, for example, uncertainty rises from 7.4 it rises from 7.0 markers per 1,000 words on clean input to 12.3–12.1 at a 25% typo rate and 18.7 at a 18.4 at 75% typo rate, when , with the real-word ratio is held at 10%. The same increasing pattern appears on ARC and trend holds on MATH-500 . Increasing the real-word ratio at and ARC at the higher typo rates, although on ARC the mildest rate stays close to clean (13.0–14.2 vs. 13.6). At a fixed typo rate leads to an additional rise in, a higher real-word ratio tends to add further uncertainty (Figure 5; Table A4).

Second-guessing shows a different pattern behaves differently. Its density increases only slightly between clean and corrupted inputs and then remains relatively stable rises only slightly on GSM8K, barely on MATH-500 and ARC, and stays roughly flat across typo rates and real-word ratios. Although the total The number of second-guessing expressions per trace increases, this is largely explained by the fact that does grow, but mostly because corrupted inputs produce

longer reasoning traces. In other words, types traces. Typos thus make the model express more uncertainty, but do not appear to make it much more uncertainty but reconsider its reasoning more often (Figure 5) only slightly more often.

The difference becomes contrast is clearer when we separate correct and incorrect answers. For the same typo rate and real-word ratio, uncertainty expressions are configuration, uncertainty is consistently more frequent in traces that end with a wrong answer. On GSM8K, uncertainty density is about 1.4 : 1.6 to 2.1 times higher in incorrect traces on GSM8K, with similar patterns on ARC and MATH-500 . Second-guessing shows a much weaker difference between correct and incorrect answers (Table A4).

Overall, uncertainty language is strongly associated with model failure, while explicit self-correction is not. The model expresses more doubt as the input becomes noisier and also tends to express more doubt when it is about to give a wrong answer. However, this relationship, while second-guessing differs much less (1.0 to 1.4 times). This relation is not unique to typo-corrupted inputs typos: even on clean GSM8K questions, incorrect answers already contain substantially more uncertainty language than correct ones.

In the LLM-as-a-judge analysis, a separate language model examined each reasoning trace and assigned a The judge’s self-doubt score from 0 to 10. A score of 0 means that the model followed a clear, straight-line solution without hesitation, while 10 means that the trace contained pervasive uncertainty, repeated second-guessing, reversals, or looping. The judgewas shown the corrupted words, the question, and the model’s reasoning trace (like in 5.3).

On score (Table 2) rises with the typo rate on all three datasets, the average self-doubt score increases as the typo rate rises, indicating that more corrupted words cause more hesitation and re-evaluation. Typo traces also and typo traces score higher than clean traces ones everywhere. The real-word ratio, by contrast, has almost no effect: at a fixed typo rate the score moves by at most about 0.1 point in either direction. The amount of corruption changes how much the model hesitates, while however, shows no consistent direction: pooled over typo rates, the score varies by less than 0.2 points on each dataset. The judge therefore confirms that more corruption

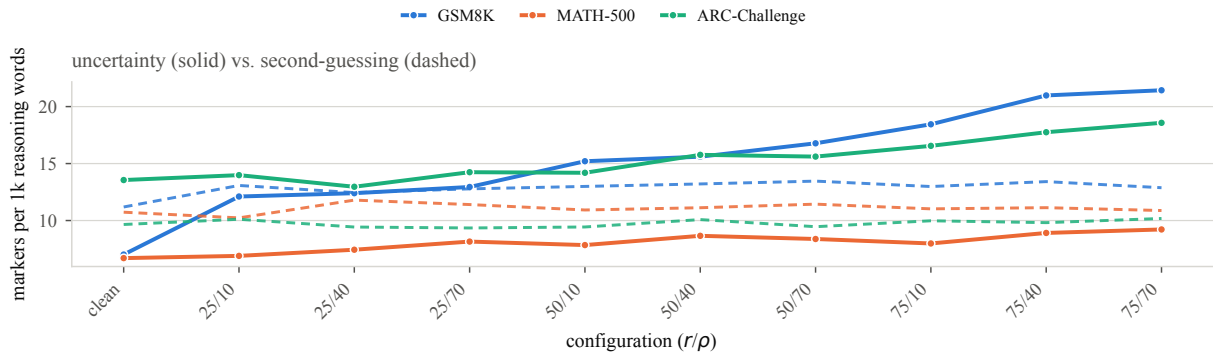


Figure 5: **Self-doubt under typos.** Marker density per 1,000 reasoning words across the grid. Uncertainty markers (solid) climb steadily rise with corruption on all three datasets (on ARC only above  $r=25$ ), most steeply on GSM8K, while second-guessing markers (dashed) stay roughly flat, so: typos make the model express much more doubt without making it but verify only slightly more. The same densities split by final answer are ratio of uncertainty density in wrong- to correct-answer traces is in Table A4.

brings more hesitation but, unlike the marker counts, does not register an additional effect of the kind of corruption does not (Table 2).

**GSM8K**      **MATH-500**      **ARC-C**  
 (lr)2-3(lr)4-5(lr)6-7 **condition** repair doubt repair  
 doubt repair doubt *clean* — 0.13 4.50 0.03 3.66  
 0.03 4.69 *by typo rate  $r$*  — 25% 0.88 4.86 0.35  
 3.90 0.97 5.12 — 50% 1.27 5.12 0.35 3.98 1.03  
 5.31 — 75% 1.74 5.64 0.48 4.01 1.35 5.63 *by*  
*real-word ratio  $\rho$*  — 10% 1.14 5.21 0.32 3.87 0.94  
 5.33 — 40% 1.32 5.29 0.39 4.03 1.10 5.27 — 70%  
 1.42 5.12 0.47 3.99 1.31 5.45 *by final answer*  
*correct* 0.87 4.69 0.27 3.75 0.88 4.94 — *wrong*  
 2.30 6.75 1.30 5.81 1.62 6.88 Mean LLM-judge  
 scores. — *repair* is the repair-understanding score  
 (0–5, higher means less of the intended meaning  
 recovered); *doubt* is the self-doubt score (0–10,  
 higher means more hesitation and looping). All  
 three datasets were judged on the same main  
 generations reported everywhere else in this  
 paper.

## 5.5 **Every No** lightweight mitigation **fails** reliably helps

We tested the three mitigation strategies proposed in our plan, all from our plan on GSM8K (Figure ??, Table 3).

**Warn.** We first added a short warning that the input Prefixing the prompt with “Note: the text may contain typos. Across the nine typo configurations, this” changed accuracy by only –0.5 –1.0 percentage points on average across the nine typo configurations, with effects ranging from –4.3 to +3.5 points. In other words, simply warning the model does not improve performance.

+1.2 points. This is consistent with the analysis in Section 5.3 §5.3, where the model often already showed signs of noticing that the input was corrupted. The main limitation therefore: the limitation does not appear to be awareness of the typo itself.

**Rewrite.** We next instructed Instructing the model to first rewrite the question with the typos corrected and only then solve it. This performed worse than the original setup prompt in all nine configurations, reducing accuracy by 4.7 5.2 percentage points on average and by as much as 7.5 up to 7.6 points. Median reasoning length also dropped from fell, from about 1,214 to about 730 tokens 300 to about 740 tokens (median over the nine typo configurations). One possible explanation is that the rewrite step forces the model to commit to a single interpretation of the corrupted question too early. If; if the reconstruction is wrong, the model then reasons about the wrong version of the problem with little opportunity to recover. The additional rewrite step may also shift part of the generation budget away from the reasoning process itself.

**Spell check.** We applied an external spellchecker spell checker (pyspellchecker, edit distance 2, prose words of four or more letters only) to the same 500 GSM8K questions before sending them to the model. A typo was considered counts as recovered only if it was is restored exactly to the original word. Recovery dropped as the share of real-word typos increased: from 54.9 55.0% at  $\rho=10$  to 39.9% at  $\rho=40$  and 25.8% at  $\rho=70$ . However, this This lexical recovery did not trans-

|       | no fix                      |      | warn                        |                             | rewrite                     |                             | spell check                 |  |
|-------|-----------------------------|------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|--|
| $r/p$ | acc                         | acc  | $\Delta$                    | acc                         | $\Delta$                    | acc                         | $\Delta$                    |  |
| 25/10 | 85.5                        | 85.7 | +0.2                        | 83.8                        | -1.7                        | 87.0                        | +1.5                        |  |
| 25/40 | 87.0                        | 87.2 | +0.2                        | 82.9                        | -4.1                        | 86.1                        | -0.9                        |  |
| 25/70 | 86.7                        | 85.5 | -1.2                        | 80.2                        | -6.6                        | 85.3                        | -1.4                        |  |
| 50/10 | 79.6                        | 79.4 | -0.2                        | <del>76.3</del> <u>76.4</u> | <del>-3.3</del> <u>-3.2</u> | 79.0                        | -0.6                        |  |
| 50/40 | 79.0                        | 74.7 | -4.3                        | 71.5                        | -7.5                        | <del>75.0</del> <u>75.1</u> | <del>-4.0</del> <u>-3.9</u> |  |
| 50/70 | 72.8                        | 74.0 | +1.2                        | 67.9                        | -4.9                        | 73.2                        | +0.4                        |  |
| 75/10 | <del>73.8</del> <u>74.2</u> | 70.6 | <del>-3.2</del> <u>-3.7</u> | 66.9                        | <del>-6.9</del> <u>-7.4</u> | 75.3                        | <del>+1.5</del> <u>+1.0</u> |  |
| 75/40 | 67.3                        | 66.3 | <del>+0.9</del> <u>-1.0</u> | 63.8                        | -3.5                        | 68.1                        | +0.9                        |  |
| 75/70 | <del>60.4</del> <u>64.5</u> | 63.9 | <del>+3.5</del> <u>-0.6</u> | 56.9                        | <del>-3.5</del> <u>-7.6</u> | 62.2                        | <del>+1.8</del> <u>-2.3</u> |  |
| mean  |                             |      | <del>-0.5</del> <u>-1.0</u> |                             | <del>-4.7</del> <u>-5.2</u> |                             | <del>-0.1</del> <u>-0.6</u> |  |

Table 3: GSM8K answered-only accuracy (%) under each prompt-level mitigation, per configuration, with the change against the unmodified prompt (no fix). The mean row is taken over the nine typo configurations.

late into a consistent accuracy gain. Across the nine typo configurations, the spellchecker: the spell checker changed accuracy by only  $-0.1$   $-0.6$  percentage points on average, improving five four conditions and hurting four; accuracy in the hardest condition was five, including the hardest one (62.2% (Table 3)).

This is likely because spellcheckers are much better at detecting against 64.5% without it). Spell checkers detect non-word errors far better than real-word substitutions, which already look like valid words. In some cases, they and may even replace a non-word typo with the a wrong valid word, effectively creating the type creating the kind of error that Section 5.2 §5.2 found to be more harmful.

Taken together, these results suggest that the main difficulty is not simply detecting that the input contains errors. None of the three strategies reliably recovers the information that was lost during corruption: the identity of the original word. Warnings provide little additional information, an explicit rewrite step can commit the model to an incorrect reconstruction, and dictionary-based correction is least effective precisely when the corruption results in another valid word.

**Prompt-level mitigations on GSM8K.** (a) answered-only accuracy for the unmodified prompt, the typo warning and the rewrite-first instruction. (b) each fix as a change from the unmodified prompt. Warning the model is null on average and the rewrite instruction is consistently harmful.

## 6 Discussion

The most consistent A recurring finding across our analyses is that real-word typos are more harm-

ful than non-word typos. This pattern appears in the per-typo regression, and in the paired comparisons, and the spellchecker results. At first, this may seem surprising, at higher typo rates, and they are also harder for the spell checker to restore. One might expect the opposite, since non-word typos often produce unusual tokenization patterns and might therefore be expected to be more disruptive disrupt tokenization. Our results suggest the opposite. A that a non-word typo is visibly unusual and can serve as a signal that something is wrong, while is more often repaired silently, whereas a real-word typo is fluent, tokenized normally, and may fit the sentence well enough to go unnoticed even when its meaning is incorrect.

This also helps explain the limited success of that its wrong meaning is taken at face value. The same property makes it hard for dictionary-based correction. Such methods are effective mainly when the corrupted form is not a valid word. Real-word substitutions, however, are already present in the dictionary and are therefore much harder to detect. In our spellchecker experiment, recovery dropped substantially as the proportion of real-word typos increased. In some cases, the spellchecker also replaced a non-word typo with an incorrect valid word, effectively turning a visible error into the type of corruption we found to be more damaging, which can only act on forms that are not valid words.

The trace analysis points to a related conclusion. The model often appears to notice notices that the input is unusual: uncertainty language increases, suspicious corrupted words are flagged more often, and reasoning traces become longer as corruption increases. However, this. This additional effort does not reliably lead to recovery. The model expresses more uncertainty, but does not appear to verify its interpretation more effectively.

This helps, which may also explain why the simple mitigation strategies we tested mitigations had little effect. Warning the model adds information it: a warning adds information the model often already has, while forcing it to rewrite the question can make it commit too a forced rewrite commits it early to a single interpretation. If that reconstruction is wrong, the subsequent reasoning is based on the wrong version of the problem.

Taken together, these results suggest that the, possibly wrong, interpretation. The main difficulty is therefore not detecting that a typo exists,

but recovering the intended word when the corruption is ambiguous. A more promising direction may ~~therefore be to preserve multiple possible interpretations be to keep multiple interpretations open~~ during reasoning and resolve ~~the ambiguity them~~ later using the structure of the problem, ~~rather than forcing an early correction.~~

~~This is consistent with the concurrent finding that a thinking budget absorbs most keyboard noise (Hu et al., 2026): in our data, too, the mildest typo rate costs little on MATH-500 and ARC. Dense corruption and real-word typos are where long reasoning stops recovering. Finally, longer reasoning-traces should not be interpreted read as evidence of successful self-correction. In our results, incorrect answers tend to: incorrect answers produce longer traces even on clean input. This suggests that additional reasoning often reflects difficulty rather than successful recovery~~

## 7 Conclusion

~~Typos have a clear effect on reasoning performance. On GSM8K, accuracy dropped by as much as 28.1 percentage points, with larger declines as the corruption rate increased, and, per typo, the strongest effect came from typos that turned one valid English word into another. The main problem is not that the model fails to notice corrupted input: it often recognizes that something is wrong, but cannot reliably recover the intended meaning. Its traces show much more uncertainty but only slightly more verification, and neither prompting strategies nor dictionary-based correction consistently improved performance. Robustness to everyday typing errors should therefore be treated as a capability to be developed explicitly, rather than assumed from strong benchmark performance.~~

~~All code is available in the project repository, and the raw generation logs are hosted on the Hugging Face Hub (Dolevabudi/silent-tax-results). All reported results can be reproduced from them with the steps in the repository README.~~

## 8 Limitations

### Limitations

**One model, one sample.** All headline results come from a single 7B reasoning model at one decoding setting with  $n=1$  sample per question. Our plan called for several samples per question, but the

API budget did not permit it, so we cannot separate typo effects from decoding variance at the level of an individual question. ~~Still, the overall effects~~ ~~The effects in the most corrupted condition~~ are much larger than expected sampling noise (McNemar  $p < 10^{-6}$  on  ~~$n=480$  paired items every dataset~~), but ~~generalisation~~ ~~generalization~~ to larger reasoning models is untested.

**Dataset substitution.** We originally planned to use GPQA-Diamond (Rein et al., 2024), but replaced it with ARC-Challenge, which is also a multiple-choice science dataset. ~~On clean GPQA-Diamond~~ ~~In a preliminary run that is not part of the released results~~, the model reached only about 52% accuracy ~~on clean GPQA-Diamond~~. Since random guessing gives 25%, this leaves limited room to measure further drops under corruption: as performance gets closer to chance, changes in accuracy become harder to interpret as reasoning failures rather than guessing. ARC-Challenge is easier and its answer options may help the model recover some information, so our multiple-choice results may underestimate the effect of typos on harder science questions.

~~Mitigations and human validation~~ ~~Scope of the mitigation study~~. We tested the three mitigation methods only on GSM8K because the experiments required additional paid API calls and our budget was limited. We chose ~~to focus on~~ GSM8K because it showed the clearest drop in answer quality under typo corruption, making it the most informative dataset for evaluating whether these mitigation strategies could recover performance.

## 8 AI Disclosure and Reflection

### AI Disclosure and Reflection

**What we used AI for.** We used GitHub Copilot with Claude primarily to support coding tasks during the project. It helped us build the typo-generation and inference pipelines, write analysis scripts, generate matplotlib code for figures, and improve the wording of some parts of the paper. All methodological and experimental decisions, including the two-axis corruption grid, typo-type weights, answer-extraction rules, and judge rubric, were made by us. The analysis scripts were written with Claude’s assistance based on these decisions, and all reported results were generated from the raw generation logs using those scripts.

**Reflection.** AI tools were especially useful for speeding up repetitive coding and analysis tasks and for keeping the implementation consistent across experiments. However, generated code and outputs still required careful checking, particularly for numerical results. We therefore verified reported results against the raw data and used a consistent scoring procedure throughout the analysis. Overall, the project highlighted the importance of reproducibility and independent verification when AI tools are used in research.

## 8 Conclusion

~~Typos have a clear and consistent effect on reasoning performance. On GSM8K, accuracy dropped by as much as 31 percentage points, with larger declines as the corruption rate increased. The strongest effect came from typos that turned one valid English word into another.~~

~~Our analysis suggests that the main problem is not that the model fails to notice corrupted input. In many cases, it appears to recognize that something is wrong, but it still cannot reliably recover the intended meaning. The reasoning traces show more uncertainty than actual verification, and the mitigation methods we tested, including prompting strategies and dictionary-based correction, did not consistently improve performance.~~

~~Overall, these results suggest that robustness to everyday typing errors should be treated as a separate capability that needs to be developed explicitly, rather than something that can be assumed from strong benchmark performance.~~

~~All code is available in the project repository, and the raw generation logs are hosted on the Hugging Face Hub (ronshtricker/typo-reasoning-results). All reported results can be reproduced by running `download_data.py` followed by `run_all.py`.~~

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## Appendix

| config        | GSM8K |       |      |      |      |      | MATH-500 |         |         |      |       |      | ARC-Challenge |       |         |      |         |       |      |      |      |      |      |       |       |       |       |       |         |       |
|---------------|-------|-------|------|------|------|------|----------|---------|---------|------|-------|------|---------------|-------|---------|------|---------|-------|------|------|------|------|------|-------|-------|-------|-------|-------|---------|-------|
|               | acc   | $n_a$ | tok  | r→w  | w→r  | $p$  | acc      | $n_a$   | tok     | r→w  | w→r   | $p$  | acc           | $n_a$ | tok     | r→w  | w→r     | $p$   |      |      |      |      |      |       |       |       |       |       |         |       |
| clean         | 91.4  | 92.6  | 500  | 1582 | 1535 | —    | —        | 95.2    | 496     | 2927 | —     | —    | —             | 84.7  | 85.2    | 485  | 494     | 1172  | 1167 | —    | —    | —    |      |       |       |       |       |       |         |       |
| typo25_real10 | 85.5  | 497   | 1912 | 1913 | 48   | 19   | 13       | 6.2e-04 | 1.3e-05 | 93.9 | 495   | 2945 | 12            | 7     | 0.359   | 86.5 | 86.2    | 487   | 492  | 1222 | 1218 | 24   | 26   | 31    | 30    | 0.418 | 0.689 |       |         |       |
| typo25_real40 | 87.0  | 500   | 1814 | 40   | 42   | 18   | 14       | 0.006   | 3.1e-04 | 93.4 | 498   | 3111 | 15            | 9     | 0.307   | 85.6 | 86.2    | 486   | 493  | 1246 | 1243 | 28   | 27   | 33    | 32    | 0.609 | 0.603 |       |         |       |
| typo25_real70 | 86.7  | 497   | 1843 | 43   | 19   | 13   | 0.003    | 1.1e-04 | 93.9    | 494  | 2992  | 15   | 9             | 0.307 | 82.0    | 82.4 | 489     | 495   | 1234 | 1239 | 42   | 41   | 26   | 0.069 | 0.087 |       |       |       |         |       |
| typo50_real10 | 79.6  | 500   | 2024 | 78   | 80   | 19   | 15       | <1e-6   | 92.8    | 498  | 2939  | 18   | 9             | 0.124 | 81.9    | 82.2 | 487     | 490   | 1298 | 1296 | 41   | 42   | 27   | 28    | 0.115 | 0.120 |       |       |         |       |
| typo50_real40 | 79.0  | 500   | 2084 | 83   | 84   | 21   | 16       | <1e-6   | 91.4    | 498  | 3085  | 22   | 6             | 0.005 | 82.5    | 82.4 | 485     | 493   | 1357 | 1359 | 42   | 28   | 25   | 0.120 | 0.051 |       |       |       |         |       |
| typo50_real70 | 72.8  | 500   | 2094 | 108  | 110  | 15   | 11       | <1e-6   | 91.5    | 496  | 3087  | 25   | 9             | 0.010 | 82.4    | 82.7 | 484     | 490   | 1329 | 1325 | 45   | 44   | 29   | 30    | 0.081 | 0.131 |       |       |         |       |
| typo75_real10 | 73.8  | 74.2  | 488  | 489  | 2084 | 2104 | 102      | 101     | 14      | 9    | <1e-6 | 89.1 | 494           | 2945  | 34      | 5    | 7.3e-06 | 80.7  | 81.0 | 488  | 490  | 1381 | 1383 | 55    | 56    | 36    | 35    | 0.059 | 0.036   |       |
| typo75_real40 | 67.3  | 495   | 2594 | 2593 | 136  | 138  | 17       | 13      | <1e-6   | 88.1 | 496   | 3125 | 39            | 7     | 4.9e-06 | 77.3 | 77.5    | 485   | 489  | 1413 | 1408 | 63   | 65   | 29    | 28    | 5.81  | 9e-04 |       |         |       |
| typo75_real70 | 60.4  | 64.5  | 480  | 482  | 2306 | 2330 | 164      | 148     | 14      | 11   | <1e-6 | 86.9 | 497           | 3138  | 3137    | 46   | 7       | <1e-6 | 74.7 | 74.5 | 486  | 490  | 1453 | 1447  | 75    | 77    | 26    | 25    | 1.8e-06 | <1e-6 |

Table A1: Every configuration on every dataset. *acc* is answered-only accuracy (%),  $n_a$  the number of questions that produced a final answer, *tok* the mean generated tokens over answered traces. r→w and w→r are paired right-to-wrong and wrong-to-right flips against the clean condition over jointly answered questions, with the continuity-corrected McNemar  $p$ -value (uncorrected for multiple comparisons).

|                                   | GSM8K              | MATH-500           | ARC-C              |
|-----------------------------------|--------------------|--------------------|--------------------|
| traces in the fit, $n$            | 4457               | 4460               | 4466               |
| intercept                         | +2.097             | 2.091              | +3.019             |
| coefficient, real-word            | -0.0761            | -0.0705            | -0.0761            |
| coefficient, non-word             | -0.0398            | -0.0402            | -0.0588            |
| odds multiplier, real-word        | 0.927              | 0.932              | 0.927              |
| odds multiplier, non-word         | 0.961              | 0.943              | 0.992              |
| odds lost per real-word typo      | 7.3                | 6.8%               | 7.3%               |
| odds lost per non-word typo       | 3.9%               | 5.7%               | 0.8%               |
| relative loss, real / non-word    | 1.9                | 1.7×               | 1.3×               |
| 95% CI (bootstrap over questions) | [1.4, 2.3]         | [1.0, 1.7]         | [52.4, 69.8]       |
| non-word coefficient, 95% CI      | [-0.0528, -0.0284] | [-0.0792, -0.0398] | [-0.0272, +0.0118] |

Table A2: Per-typo logistic fit behind [Section 5.2§5.2](#). One regression per dataset, correct  $\sim$  num\_real + num\_nonword, fitted over all answered typo traces pooled across the nine configurations. The predictors are the counts of each kind of typo actually applied to that question. An odds multiplier is  $\exp(\text{coefficient})$ : the factor by which one additional typo of that kind multiplies the odds of a correct answer. Confidence intervals come from 1,000 bootstrap resamples of questions.

| $r$ | $\rho$ | GSM8K                          |                                |                                    | MATH-500           |       |                               | ARC-Challenge                |                               |                               |
|-----|--------|--------------------------------|--------------------------------|------------------------------------|--------------------|-------|-------------------------------|------------------------------|-------------------------------|-------------------------------|
|     |        | $\Delta\text{acc}$             | r/w                            | $p$                                | $\Delta\text{acc}$ | r/w   | $p$                           | $\Delta\text{acc}$           | r/w                           | $p$                           |
| 25  | 10→70  | +1.0                           | 31/36                          | 0.625                              | +0.0               | 10/10 | <del>0.823</del> <u>1.000</u> | <del>-3.8*</del> <u>-3.1</u> | <del>46/28</del> <u>45/30</u> | <del>0.048</del> <u>0.106</u> |
| 25  | 10→40  | +1.4                           | 28/35                          | 0.450                              | +0.0               | 10/10 | <del>0.823</del> <u>1.000</u> | <del>-0.4</del> <u>+0.2</u>  | <del>32/30</del> <u>33</u>    | <del>0.899</del> <u>1.000</u> |
| 25  | 40→70  | -0.4                           | 30/28                          | 0.896                              | -0.2               | 13/12 | 1.000                         | <del>-3.8</del> <u>4.1*</u>  | <del>40/22</del> <u>41/21</u> | <del>0.031</del> <u>0.016</u> |
| 50  | 10→70  | -6.8*                          | 76/42                          | 0.002                              | -1.4               | 19/12 | 0.281                         | <del>+0.6</del> <u>0.4</u>   | <del>38/41</del> <u>43</u>    | <del>0.822</del> <u>0.913</u> |
| 50  | 10→40  | -0.6                           | 45/42                          | 0.830                              | -1.4               | 17/10 | 0.248                         | <del>+0.0</del> <u>0.2</u>   | <del>35/35</del> <u>36/37</u> | <del>0.905</del> <u>1.000</u> |
| 50  | 40→70  | -6.2*                          | 70/39                          | 0.004                              | -0.2               | 13/12 | 1.000                         | <del>-0.9</del> <u>+0.2</u>  | <del>39/35</del> <u>37/38</u> | <del>0.727</del> <u>1.000</u> |
| 75  | 10→70  | <del>-14.1</del> <u>-11.0*</u> | <del>109/43</del> <u>94/42</u> | <del>&lt;1e-6</del> <u>1.2e-05</u> | -1.8               | 28/19 | 0.243                         | <del>-6.7</del> <u>6.9*</u>  | <del>60/61</del> <u>28</u>    | <del>9.56</del> <u>9e-04</u>  |
| 75  | 10→40  | <del>-5.8</del> <u>6.2*</u>    | <del>85/87</del> <u>57</u>     | <del>0.023</del> <u>0.016</u>      | -0.2               | 20/19 | 1.000                         | -4.0                         | <del>54/35</del> <u>52/33</u> | <del>0.056</del> <u>0.051</u> |
| 75  | 40→70  | <del>-7.8*</del> <u>-3.6</u>   | <del>93/56</del> <u>74/57</u>  | <del>0.003</del> <u>0.162</u>      | -1.6               | 25/17 | 0.280                         | <del>-3.4</del> <u>-3.1</u>  | <del>60/44</del> <u>45</u>    | <del>0.141</del> <u>0.172</u> |

Table A3: Controlled paired contrast behind [Section 5.2§5.2](#), at every typo rate and every pair of real-word ratios. Within a fixed  $r$  the same words of the same questions are corrupted in the same positions, so only the *kind* of corruption changes.  $\Delta\text{acc}$  is the accuracy of the higher- $\rho$  variant minus the lower, over questions answered in both; r/w is right-to-wrong against wrong-to-right flips;  $p$  is a continuity-corrected McNemar test and \* marks  $p < 0.05$  before multiple-comparison correction.

| config        | GSM8K     |                  |                  |           |           |           | MATH-500  |                  |                  |         |           |           | ARC-Challenge |                  |                  |           |           |           |
|---------------|-----------|------------------|------------------|-----------|-----------|-----------|-----------|------------------|------------------|---------|-----------|-----------|---------------|------------------|------------------|-----------|-----------|-----------|
|               | tok       | tok <sub>c</sub> | tok <sub>w</sub> | uncert.   | 2nd-g.    | u.w/c     | tok       | tok <sub>c</sub> | tok <sub>w</sub> | uncert. | 2nd-g.    | u.w/c     | tok           | tok <sub>c</sub> | tok <sub>w</sub> | uncert.   | 2nd-g.    | u.w/c     |
| clean         | 4582-1535 | 4406-1412        | 3456-3082        | 7.4-7.0   | 43.0-11.2 | 4.72-1.59 | 2927      | 2803             | 5352             | 6.8-6.7 | 42.6-10.7 | 1.70      | 4472-1167     | 4025-1018        | 4989-2026        | 44.0-13.6 | 44.3-9.7  | 4.66-1.71 |
| typo25_real10 | 4912-1913 | 1511             | 4281             | 42.3-12.1 | 44.7-13.1 | 1.83      | 2945      | 2826             | 4783             | 7.0-6.9 | 44.8-10.2 | 4.63-1.61 | 4222-1218     | 4430-1118        | 4870-1837        | 44.5-14.0 | 44.8-10.1 | 1.57      |
| typo25_real40 | 1814      | 1567             | 3470-3471        | 43.6-12.4 | 44.0-12.5 | 1.90      | 3111      | 2900             | 6075             | 7.5-7.4 | 43.7-11.8 | 1.52      | 4246-1243     | 4444-1111        | 2026-2069        | 43.4-13.0 | 44.0-9.4  | 4.71-1.72 |
| typo25_real70 | 1843      | 1537             | 3843             | 43.2-12.9 | 44.5-12.8 | 2.12      | 2992      | 2859             | 5047             | 8.2     | 43.2-11.4 | 4.68-1.66 | 4234-1239     | 4066-1073        | 4998-2015        | 44.6-14.2 | 40.8-9.3  | 4.04-1.91 |
| typo50_real10 | 2024      | 1554             | 3856             | 45.4-15.2 | 44.6-13.0 | 1.87      | 2939      | 2763             | 5191             | 7.9-7.8 | 42.7-10.9 | 1.97      | 4208-1296     | 4484-1182        | 4844-1828        | 44.7-14.2 | 44.0-9.4  | 4.50-1.51 |
| typo50_real40 | 2084      | 1631             | 3787             | 45.9-15.6 | 44.7-13.2 | 4.63-1.64 | 3085      | 2839             | 5694             | 8.8-8.7 | 42.7-11.1 | 1.96      | 4357-1359     | 4206-1214        | 2060-2037        | 46.3-15.8 | 44.7-10.1 | 4.92-1.91 |
| typo50_real70 | 2094      | 1501             | 3682             | 47.1-16.8 | 44.8-13.5 | 4.65-1.66 | 3087      | 2854             | 5600             | 8.5-8.4 | 43.1-11.4 | 1.90      | 4329-1325     | 4480-1176        | 2022-2033        | 46.1-15.6 | 44.1-9.5  | 4.67-1.66 |
| typo75_real10 | 2084-2104 | 4588-1598        | 3479-3559        | 48.7-18.4 | 44.4-13.0 | 4.62-1.60 | 2945      | 2787             | 4236             | 8.1-8.0 | 42.7-11.0 | 2.06-2.05 | 4381-1383     | 4232-1236        | 2006-2012        | 47.1-16.6 | 44.5-10.0 | 1.48      |
| typo75_real40 | 2594-2593 | 1786             | 4253             | 41.3-21.0 | 44.7-13.4 | 1.60      | 3125      | 2881             | 4932             | 9.0-8.9 | 42.7-11.1 | 2.06      | 4413-1408     | 4263-1260        | 4925-1919        | 48.4-17.8 | 44.3-9.8  | 4.38-1.40 |
| typo75_real70 | 2306-2330 | 4652-1638        | 3306-3588        | 41.9-21.4 | 44.1-12.9 | 4.55-1.63 | 2438-3137 | 2042-2911        | 4640             | 9.3-9.2 | 42.5-10.9 | 1.75      | 4453-1447     | 4263-1260        | 2045-1995        | 49.2-18.6 | 44.6-10.2 | 1.47      |

Table A4: Supporting numbers for [Sections 5.1 §5.1](#) and [5.4 §5.4](#). *tok* is mean generated tokens over all answered traces; *tok<sub>c</sub>* and *tok<sub>w</sub>* are the same mean ~~taken~~ over traces that ended in a correct and in a wrong answer. *uncert.* and *2nd-g.* are uncertainty and second-guessing markers per 1,000 reasoning words, and *u.w/c* is uncertainty density in wrong-answer traces divided by correct-answer traces. Traces [generally](#) lengthen with the typo rate ~~on every~~ [dataset](#) ([MATH-500 at  \$\rho=10\$  and  \$\rho=40\$  is flat](#)), a wrong answer is preceded by a longer trace than a correct one in every configuration, and uncertainty climbs while second-guessing stays [roughly](#) flat.

| r  | $\rho$ | GSM8K     |           |         |           |                  |                  | MATH-500  |           |         |           |                  |                  | ARC-Challenge |           |         |           |                  |                  |
|----|--------|-----------|-----------|---------|-----------|------------------|------------------|-----------|-----------|---------|-----------|------------------|------------------|---------------|-----------|---------|-----------|------------------|------------------|
|    |        | silent    | flag      | mis.    | unused    | mis <sub>c</sub> | mis <sub>w</sub> | silent    | flag      | mis.    | unused    | mis <sub>c</sub> | mis <sub>w</sub> | silent        | flag      | mis.    | unused    | mis <sub>c</sub> | mis <sub>w</sub> |
| 25 | 10     | 73.3-74.0 | 41.9-11.3 | 2.9     | 42.0-11.9 | 2.5              | 5.0              | 78.0-77.7 | 2.3-1.7   | 1.2     | 48.4-19.3 | 1.3              | 0.4              | 80.0-80.8     | 5.8-5.2   | 2.1-1.9 | 42.1-12.0 | 4.8-1.6          | 3.9-3.8          |
| 25 | 40     | 70.2-72.4 | 45.1-12.9 | 4.3-4.1 | 40.4-10.5 | 3.9-3.8          | 6.8-6.3          | 73.1-74.5 | 5.8-3.3   | 4.6-1.5 | 49.5-20.7 | 4.3-1.2          | 3.9-3.4          | 74.8-76.9     | 9.1-7.0   | 3.1-2.6 | 43.1-13.5 | 2.7-2.1          | 5.2-5.1          |
| 25 | 70     | 64.6-68.1 | 49.3-15.4 | 4.8-4.7 | 41.3-11.8 | 4.3-4.2          | 7.7-7.8          | 71.4-75.0 | 40.2-5.6  | 4.7-1.5 | 46.8-17.9 | 4.1-1.3          | 4.1-3.4          | 69.6-73.3     | 42.2-9.1  | 5.3-4.1 | 42.9-13.4 | 5.0-3.7          | 6.4-5.7          |
| 50 | 10     | 65.8-66.3 | 48.7-18.4 | 4.7-4.6 | 40.9-10.7 | 3.9-3.7          | 7.7-7.5          | 75.0-75.1 | 4.3-3.6   | 1.4     | 49.4-19.9 | 4.3-1.2          | 2.5              | 76.9-77.5     | 8.1-7.6   | 2.5-2.4 | 12.5      | 2.0-1.8          | 4.9              |
| 50 | 40     | 60.6-62.6 | 23.4-21.4 | 5.2-5.0 | 40.8-11.0 | 5.1-4.8          | 5.8-5.7          | 70.2-71.2 | 8.3-6.0   | 2.3-2.1 | 49.2-20.7 | 2.2-2.0          | 3.2-2.7          | 71.8-74.0     | 41.7-9.6  | 3.7-3.4 | 42.7-13.1 | 3.4-3.0          | 5.1-4.7          |
| 50 | 70     | 55.9-59.0 | 25.3-22.0 | 6.6-6.3 | 42.2-12.7 | 6.1-5.9          | 7.8-7.3          | 67.2-70.5 | 41.3-6.9  | 2.9-2.6 | 48.6-20.0 | 2.4-2.1          | 6.0-5.3          | 66.7-70.3     | 44.2-10.9 | 5.4-4.7 | 43.7-14.1 | 5.1-4.5          | 6.2-5.6          |
| 75 | 10     | 55.8      | 26.1-26.9 | 6.9-6.5 | 41.3-10.8 | 6.1-5.8          | 8.7-8.2          | 70.4-70.6 | 8.2-7.6   | 2.1-2.0 | 49.3-19.8 | 4.7-1.6          | 3.7-3.4          | 71.0-71.9     | 43.4-12.7 | 3.4-3.2 | 42.2-12.1 | 2.9-2.8          | 5.2-5.0          |
| 75 | 40     | 47.4-48.8 | 32.9-32.3 | 8.3-7.8 | 41.4-11.2 | 6.8-6.1          | 41.1-10.7        | 68.4-69.9 | 9.6-7.3   | 2.4-2.3 | 49.5-20.5 | 4.9-1.6          | 4.8              | 65.8-67.9     | 46.3-14.5 | 5.0-4.5 | 42.8-13.1 | 4.7-4.1          | 6.2-5.8          |
| 75 | 70     | 46.2-48.3 | 30.9-29.4 | 9.6-9.2 | 43.3-13.0 | 8.0-7.8          | 41.9-11.5        | 62.2-65.1 | 15.2-11.0 | 4.0-3.7 | 48.5-20.2 | 3.5-3.1          | 6.0-5.7          | 59.4-62.5     | 20.3-17.2 | 6.9-6.2 | 43.3-14.0 | 6.1-5.3          | 9.3-8.9          |

Table A5: Every corrupted word [of three or more letters](#) in every [answered question of every](#) configuration, classified by how the reasoning trace handles it, as a percentage of the corrupted words in that configuration; [words whose corrupted form also appears in the model's trace on the clean question \(about 3% overall, more at higher  \$\rho\$ \) are left out.](#) *silent* / *flag* / *mis.* / *unused* are the four categories of [Section 5.3 §5.3](#). *mis<sub>c</sub>* and *mis<sub>w</sub>* are the misread rate among questions the model answered correctly and among those it answered wrongly, which is the comparison behind the claim that misreading is associated with failure. Figure 4 shows the same data pooled along each axis.